AQUÍ VA EL TÍTULO DEL ARTÍCULO

ABSTRACT. Aquí va el resumen

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[PP21, SS06a, Hoc69]

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[Sim06a, Sex03, Joh82]

3. COMPACT QUOTIENTS

Let $f\colon A \to B$ be a frame morphism with right adjoint $f_*\colon B \to A$. If $F \subseteq A$ and $G \subseteq B$ are filters, we define:

$$b \in f^*F \iff f_*(b) \in F \qquad a \in f_*G \iff f(a) \in G, \tag{1}$$

where f^*F and f_*G are filters in B and A, respectively.

Proposition 3.1. *For $f = f^*\colon A \to B$ a frame morphism and $G \in B^\wedge$, then $f_*G \in A^\wedge$.*

Proof. By (1), f_*G is a filter in A. We need that f_*G satisfy the open filter condition. Let $X \subseteq A$ be such that $\bigvee X \in f_*G$, with X directed. Then

$$Y = \{f(x) \mid x \in X\}$$

is directed and $f(\bigvee X) = \bigvee f[X] = \bigvee Y \in G$. Since G is an open filter and $f(\bigvee X) = \bigvee f[X] \in G$, there exists $x \in X$ such that $f(x) \in G$. Therefore, $x \in f_* G$. □

Simmons shows in [Sim04, Lemma 8.9 and Corollary 8.10] that the following diagram

$$\begin{tikzcd} A \arrow[r, "f^\infty"] \arrow[d, "U_A"'] & A \arrow[d, "U_A"] \\ \mathcal{O}S \arrow[r, "F^\infty"'] & \mathcal{O}S \end{tikzcd} \tag{2}$$

commutes laxly, that is, $U_A \circ f^\infty \leq F^\infty \circ U_A$. In this diagram f^∞ and F^∞ represent the fitted nuclei associated with the filters $F \in A^\wedge$ and $\nabla \in \mathcal{O}S^\wedge$, respectively, that is,

$$f^\infty = v_F \quad \text{and} \quad F^\infty = v_\nabla.$$

Here we prove something more general, since we consider $j \in NA$, $F \in A_j^\wedge$ and $j_* F \in A^\wedge$ to produce the diagram

$$\begin{tikzcd} A \arrow[r, "f^\infty"] \arrow[d, "j^*"'] & A \arrow[d, "j^*"] \\ A_j \arrow[r, "\hat{f}^\infty"'] & A_j \end{tikzcd}$$

where $\hat{f}^\infty$ and f^∞ are the fitted nuclei associated with the filters $j_* F$ and F, respectively.

The following lemma is part of the analysis developed in this work and constitutes a key tool in the construction of the commutative diagrams that follow.

Lemma 3.2. *For $j = j^*$, f and $\hat{f}$ as before, it holds that $j \circ \hat{f} \leq f \circ j$.*

Proof. Recall that, by (1)

$$\hat{f} = \dot{\bigvee}\{v_y \mid y \in j_* F\} \quad \text{y} \quad f = \dot{\bigvee}\{v_{j(y)} \mid j(y) \in F\}.$$

Let $a \in A$ be. For every $y \in j_* F$, we have

$$j(v_y(a)) = j(y \succ a) \leq (j(y) \succ j(a)) = v_{j(y)}(j(a)) \leq f(j(a)),$$

where the last inequality follows from $j(y) \in F$. Therefore, since j preserves arbitrary joins,

$$j(\hat{f}(a)) = j\Big(\bigvee\{v_y(a) \mid y \in j_* F\}\Big) = \bigvee\{j(v_y(a)) \mid y \in j_* F\} \leq f(j(a)).$$

Hence, $j \circ \hat{f} \leq f \circ j$. □

We now prove the above for an arbitrary ordinal.

Corollary 3.3. *For j, f and $\hat{f}$ as before, it holds that $j \circ \hat{f}^\alpha \leq f^\alpha \circ j$ where $\alpha \in \mathrm{Ord}$.*

Proof. We proceed by transfinite induction on α If $\alpha = 0$, the result is immediate.

For the induction step, suppose that the inequality holds for α. Then

$$j \circ \hat{f}^{\alpha+1} = j \circ \hat{f} \circ \hat{f}^{\alpha} \leq f \circ j \circ \hat{f}^{\alpha} \leq f \circ f^{\alpha} \circ j = f^{\alpha+1} \circ j,$$

where the first inequality follows from Lemma 3.2 and the second from the induction hypothesis.

Finally, let λ be a limit ordinal. For every $a \in A$,

$$j(\hat{f}(a)) = j\Big(\bigvee_{\alpha<\lambda} \hat{f}^{\alpha}\Big) = \bigvee_{\alpha<\lambda} j(\hat{f}^{\alpha}(a)) \leq \bigvee_{\alpha<\lambda} f^{\alpha}(j(a)) = f^{\lambda}(j(a)).$$

where the inequality follows from the induction hypothesis. Hence

$$j \circ \hat{f}^{\lambda} \leq f^{\lambda} \circ j.$$

This completes the proof. □

Recalling that $v_F = f^{\infty}$ and $v_{j_*F} = \hat{f}^{\infty}$, Corollary 3.3 yields

$$j \circ v_{j_*F} \leq v_F \circ j$$

We now compare the compact quotients induced by F and j_*F, and construct a frame morphism between them making the corresponding diagram commute.

If we consider j, v_F and v_{j_*F} we have the following diagram

$$\begin{tikzcd} A \arrow[r, shift left, "(v_{j_*F})^*"] \arrow[d, "j"'] \arrow[dr, "H"] & A_{j_*F} \arrow[l, shift left, "(v_{j_*F})_*"] \\ A_j \arrow[r, shift right, "(v_F)^*"'] & A_F \arrow[l, shift right, "(v_F)_*"'] \end{tikzcd} \tag{3}$$

where A_F and A_{j_*F} are the compact quotients produced by v_F and v_{j_*F}, respectively.

Define

$$H\colon A \to A_F, \qquad H = v_F \circ j.$$

Since v_F is idempotent, $H(A) \subseteq A_F$. Hence, by restricting H to A_{j_*F}, we obtain a map

$$h\colon A_{j_*F} \to A_F, \qquad h = H|_{A_{j_*F}}.$$

As finite meets in quotient frames are inherited from the ambient frame, h preserves finite meets. It remains to verify that h preserves arbitrary joins.

Lemma 3.4. *The map $h\colon A_{j_*F} \to A_F$ preserves arbitrary joins.*

Proof. Let $X \subseteq A_{j_*F}$ be. Since joins in A_{j_*F} are given by

$$\bigvee_{v_{j_*F}} X = v_{j_*F}(\bigvee X)$$

we have

$$\begin{aligned} h\Big(\bigvee_{v_{j_*F}} X\Big) &= v_F\Big(j\Big(v_{j_*F}\Big(\bigvee X\Big)\Big)\Big) \\ &\leq v_F\Big(v_F\Big(j\Big(\bigvee X\Big)\Big)\Big) \\ &= v_F\Big(j\Big(\bigvee X\Big)\Big), \end{aligned}$$

where the inequality follows from Corollary 3.3 and the equality from the idempotency of v_F.

Since H is a frame morphism,

$$v_F\Big(j\Big(\bigvee X\Big)\Big) = \bigvee_{v_F}\{H(x) \mid x \in X\} = \bigvee_{v_F}\{h(x) \mid x \in X\}.$$

Therefore,

$$h\Big(\bigvee_{v_{j_*F}} X\Big) \leq \bigvee_{v_F} h[X].$$

Conversely, for every $x \in X$,

$$x \leq \bigvee_{v_{j_*F}} X,$$

and hence, by monotonicity of h,

$$h(x) \leq h\Big(\bigvee_{v_{j_*F}} X\Big).$$

Since this holds for every $x \in X$, we have

$$\bigvee_{v_F} h[X] \leq h\Big(\bigvee_{v_{j_*F}} X\Big).$$

Thus,

$$h\Big(\bigvee_{v_{j_*F}} X\Big) = \bigvee_{v_F} h[X].$$

Therefore h preserves arbitrary joins. □

The previous construction yields the following commutative diagram induced by the compact quotients associated with the fitted nuclei.

Proposition 3.5. *The diagram*

$$
\begin{tikzcd}
A \arrow[r, "v_{j_* F}"] \arrow[d, "j"'] & A_{j_* F} \arrow[d, "h"] \\
A_j \arrow[r, "v_F"'] & A_F
\end{tikzcd}
\tag{4}
$$

is commutative.

Proof. By Lemma 3.4, h is a frame morphism. The commutativity of the diagram follows from the definition of h. □

A particular instance of diagram (4) is obtained by taking $j^* = \mathcal{U}_A$, where $S = \operatorname{pt} A$. Thus, we obtain the following commutative diagram:

$$
\begin{tikzcd}
A \arrow[r, "v_F"] \arrow[d, "\mathcal{U}_A"'] & A_F \arrow[d, "h"] \\
\mathcal{O}S \arrow[r, "v_\nabla"'] & \mathcal{O}S_\nabla
\end{tikzcd}
$$

Recall that $\operatorname{pt} A_F = Q$. Hence, the spatial reflection of A_F is given by

$$\mathcal{U}_{A_F} \colon A_F \to \mathcal{O} \operatorname{pt} A_F = \mathcal{O}Q.$$

Similarly, since $\operatorname{pt}((\mathcal{O}S)_\nabla) = Q$, we have the spatial reflection

$$\mathcal{U}_{\mathcal{O}S_\nabla} \colon \mathcal{O}S_\nabla \to \mathcal{O} \operatorname{pt}(\mathcal{O}S)_\nabla = \mathcal{O}Q.$$

Consequently, the two induced morphisms from A to $\mathcal{O}Q$ are both given by

$$x \mapsto \mathcal{U}_A(x) \cap Q.$$

Therefore, the preceding constructions fit into the following commutative diagram:

$$
\begin{tikzcd}
A \arrow[r, "v_F"] \arrow[dd, "\mathcal{U}_A"'] & A_F \arrow[dd, "h"] \arrow[dr, "\mathcal{U}_{A_F}"] & \\
& & \mathcal{O}Q \\
\mathcal{O}S \arrow[r, "v_\nabla"'] & \mathcal{O}S_\nabla \arrow[ur, "\mathcal{U}_{\mathcal{O}S_\nabla}"'] &
\end{tikzcd}
\tag{5}
$$

Since the above diagram is determined by $F \in A^\wedge$ and the associated compact saturated subset $Q \in \mathcal{Q}S$, we refer to diagram (5) as the *Q-square*.

Theorem 3.6. *Under the hypotheses of the Q-square, if A satisfies property* $(\mathbf{H})$*, then*

$$\mathcal{O}Q \cong \uparrow Q'.$$

Equivalently, the frame of open sets of the point space of A_F is isomorphic to a compact closed quotient of a Hausdorff space.

Proof. Suppose that A satisfies $(\mathbf{H})$. Since $(\mathbf{H})$ is conservative, $S = \mathrm{pt}\, A$ is a T_2 space. Hence, for every $Q \in \mathcal{Q}S$, the set Q is closed, and therefore $Q' \in \mathcal{O}S$

Moreover, since every T_2 space is strongly stacked, we have

$$v_\nabla = [Q'].$$

As $Q' \in \mathcal{O}S$, it follows that

$$[Q'] = u_{Q'}$$

Consequently,

$$(\mathcal{O}S)_{v_\nabla} = (\mathcal{O}S)_{u_{Q'}} \cong \uparrow Q'.$$

Now, $u_{Q'}$ is a complemented nucleus in $\mathcal{O}S$, by [PP12, Proposition VI.3.3], implies that $\mathcal{O}S_{v_\nabla}$ is spatial. Therefore,

$$(\mathcal{O}S)_{v_\nabla} \cong \mathcal{O}\,\mathrm{pt}((\mathcal{O}S)_{v_\nabla}).$$

Since

$$\mathrm{pt}((\mathcal{O}S)_{v_\nabla}) = Q,$$

we obtain

$$\mathcal{O}Q \cong (\mathcal{O}S)_{v_\nabla} \cong \uparrow Q'.$$

□

4. Tidy frames

Proposition 4.1. *If A is a tidy frame, then A_j is a tidy frame.*

Proof. Let us first observe that $F \subseteq j_* F$. Since A is tidy and $F \in A^\wedge$, it holds that

$$x \in F \Rightarrow \hat{d} \vee x = 1,$$

where $\hat{d} = d(\alpha) = f^\alpha(0)$.

If $\hat{d} \le d$, then $d \vee x = 1$, for $d = d(\alpha) = f^\alpha(j(0))$.

Thus, by the Corollary 3.3

$$\hat{d} = \hat{d}(\alpha) \le j(\hat{d}(\alpha)) = j(\hat{f}^\alpha(0)) \le f^\alpha(j(0)) = d(\alpha) = d.$$

Therefore, if $x \in F$, then $d \vee x = 1$ and A_j is tidy. □

The following theorem establishes that tidiness is preserved under binary coproducts.

Proposition 4.2. *If $A_1, A_2 \in \mathrm{Frm}$ are tidy frames, then $A_1 \oplus A_2$ is a tidy frame.*

Proof. Let A_1, A_2 be α-tidy frames. By construction of the coproduct $A_1 \oplus A_2$, we have the following diagram

$$\begin{tikzcd} A_i \arrow[r, "\alpha_i"] \arrow[rrr, bend left, "\iota_i"] & A_1 \times A_2 \arrow[r, "\downarrow(\text{-})"] & \mathcal{L}(A_1 \times A_2) \arrow[r, "\sim"] & A_1 \oplus A_2 \arrow[lll, bend left, "(\iota_i)_*"] \end{tikzcd}$$

where $\alpha_1(a) = (a, 1)$ and $\alpha_2(b) = (1, b)$ to $a \in A_1$ and $b \in A_2$. Furthermore, as $\iota_i \in \mathrm{Frm}$, $(\iota_i)_*$ is its right adjoint.

Take $F \in (A_1 \oplus A_2)^\wedge$ and, by Proposition 3.1, $(\iota_i)_*[F] \in A_i^\wedge$. As A_i is tidy, for $x_i \in (\iota_i)_* F$ it holds that $d_i \vee x_i = 1$, where

$$d_i = d_i(\alpha) = f_i^\alpha(0), \text{ con } f_i = \dot{\bigvee}\{v_y \mid y \in (\iota_i)_* F\}$$

By (1) we have that if $x_i \in (\iota_i)_* F$, then $\iota_i(x_i) \in F$, that is,

$$x_1 \oplus 1, 1 \oplus x_2 \in F$$

and because F is a filter, we have that $x_1 \oplus 1 \wedge 1 \oplus x_2 = x_1 \oplus x_2 \in F$.

Additionally, for each $i = 1, 2$ we have the following diagram

$$\begin{tikzcd} A_i \arrow[r, "f_i"] \arrow[d, "\iota_i"'] & A_i \arrow[d, "\iota_i"] \\ A_1 \oplus A_2 \arrow[r, "f_F"'] & A_1 \oplus A_2 \end{tikzcd}$$

where $f_F = \dot{\bigvee}\{v_{\iota_i(y)} \mid \iota_i(y) \in F\}$.

Let $a \in A_i$ be, then $v_y(a) \le f_i(a)$ and $\iota_i(v_y(a)) \le \iota_i(f_i(a))$. Additionally,

$$\begin{aligned} \iota_i((y \succ a) \wedge y) = \iota_i(y \wedge a) \le \iota_i(a) &\Leftrightarrow \iota_i(y \succ a) \wedge \iota_i(y) \le \iota_i(a) \\ &\Leftrightarrow \iota_i(y \succ a) \le (\iota_i(y) \succ \iota_i(a)). \end{aligned}$$

Next

$$\iota_i(f_i(a)) \le (\iota_1(y) \succ \iota_i(a)) = v_{\iota_i(y)}(\iota_i(a)) \le f_F(\iota_i(a)).$$

Therefore, $\iota_i \circ f_i \le f_F \circ \iota_i$.

We prove by transfinite induction that $\iota_i \circ f_i^\alpha \le f_F^\alpha \circ \iota_i$ with $\alpha \in \mathrm{Ord}$.

- If $\alpha = 0$, it is trivial.
- For the induction step, assume that for α the inequality holds. Then

$$\iota_i \circ f_i^{\alpha+1} = \iota_i \circ f_i \circ f_i^\alpha \le f_F \circ \iota_i \circ f_i^\alpha \le f_F \circ f_F^\alpha \circ \iota_i = f_F^{\alpha+1} \circ \iota_i.$$

- If λ is a limit ordinal, then
$$f_i^\lambda = \bigvee\{f_i^\alpha \mid \alpha < \lambda\}, \quad f_F^\lambda = \bigvee\{f_F^\alpha \mid \alpha < \lambda\}$$
and
$$\iota_i \circ f_i^\lambda = \iota_i \circ \bigvee_{\alpha<\lambda} f_i^\alpha \leq \bigvee_{\alpha<\lambda} \iota_i \circ f_i^\alpha.$$
By the induction hypothesis we have that
$$\iota_i \circ f_i^\alpha \leq f_F^\alpha \circ \iota_i \Rightarrow \bigvee_{\alpha<\lambda} \iota_i \circ f_i^\alpha \leq \bigvee_{\alpha<\lambda} f_F^\alpha \circ \iota_i.$$

Evaluating at $0 \in A_i$, we obtain that
$$(\iota_i \circ f_i^\alpha)(0) \leq (f_F^\alpha \circ \iota_i)(0) \Rightarrow \iota_1(d_1) \leq f_F^\alpha(0 \oplus 1) \text{ y } \iota_2(d_2) \leq f_F^\alpha(1 \oplus 0).$$
Thus
$$d_1 \oplus d_2 = d_1 \oplus 1 \wedge 1 \oplus d_2 \leq f_F^\alpha(0 \oplus 1) \wedge f_F^\alpha(1 \oplus 0) = f_F^\alpha(0 \oplus 0).$$
For $d = d(\alpha) = f_F^\alpha(0 \oplus 0)$,
$$(x_1 \oplus x_2) \vee d_1 \oplus d_2 \leq (x_1 \oplus x_2) \vee d,$$
but $x_1 \oplus x_2 \vee d_1 \oplus d_2 = (x_1 \vee d_1) \oplus (x_2 \vee d_2) = 1 \oplus 1$.

Therefore,
$$x_1 \oplus x_2 \vee d = 1 \oplus 1,$$
that is, $A_1 \oplus A_2$ is tidy. □

Corollary 4.3. *If $A_1, A_2 \in \mathrm{Frm}$ are α-tidy and β-tidy, respectly, then $A_1 \oplus A_2$ is tidy.*

Proof. Let $\gamma = \max\{\alpha, \beta\}$ be, then A_1 and A_2 are γ-tidy and by Proposition 4.2, $A_1 \oplus A_2$ is tidy. □

We extend the previous result to arbitrary coproducts

Proposition 4.4. *Tidiness is preserved under coproducts.*

Proof. Let $\{A_i\}_{i\in I}$ be a family of tidy frames and $\{\alpha_i\}_{i\in I} \subseteq \mathrm{Ord}$ the efficiency degrees of each A_i. We know that
$$\bigoplus_{i\in I} A_i \in \mathrm{Frm} \quad \text{ and } \quad (\iota_i\colon A_i \to \bigoplus_{i\in I} A_i) \in \mathrm{Frm},$$
then $(\iota_i)_*$ is the right adjoint of ι_i. Additionally, for $F \in (\bigoplus_{i\in I} A_i)^\wedge$, by Proposition 3.1, $(\iota_i)_*F \in A_i^\wedge$ and by (1),
$$x_i \in (\iota_i)_*F \Leftrightarrow \iota_i(x_i) \in F.$$
Take $\alpha = \sup\{\alpha_i \mid i \in I\}$, then each A_i is α-tidy. By hypothesis, for $x_i \in (\iota_i)_*F$ it holds that $x_i \vee d_i = 1$, where
$$d_i = d_i(\alpha) = f_i^\alpha(0), \text{ with } f_i = \dot{\bigvee}\{v_y \mid y \in (\iota_i)_*F\}.$$

Claim: For all $x_i \in (\iota_i)_* F$, $\langle x_i \rangle = \bigoplus \iota_i(x_i) \in F$.

We know that $\bigoplus \iota_i(x_i) = \bigvee \iota_i(x_i)$. Then, $\iota_i(x_i) \leq \bigvee \iota_j(x_j)$ for all $i \in I$ and $j \in I$. Therefore, because F is a filter, we have that $\bigvee \iota_i(x_i) \in F$.

For each A_i there exists the following diagram

$$\begin{tikzcd} A_i \arrow[r, "f_i"] \arrow[d, "\iota_i"'] & A_i \arrow[d, "\iota_i"] \\ \bigoplus_{i\in I} A_i \arrow[r, "f_F"'] & \bigoplus_{i\in I} A_i \end{tikzcd}$$

where $f_F = \dot{\bigvee}\{v_{\iota_i(y)} \mid \iota_i(y) \in F\}$.

Proceeding as in the proof of Proposition 4.2, we obtain

$$\iota_i \circ f_i \leq f_F \circ \iota_i$$

and, by transfinite induction,

$$\iota_i \circ f_i^\alpha \leq f_F^\alpha \circ \iota_i.$$

Evaluating at $0 \in A_i$, we obtain

$$(\iota_i \circ f_i^\alpha)(0) \leq (f_F^\alpha \circ \iota_i)(0) \iff \iota_i(d_i) \leq f_F^\alpha(\langle 0 \rangle = d(\alpha) = d.$$

and $\langle d_i \rangle = \bigoplus \iota_i(d_i) \leq d$.

Thus, $\langle x_i \rangle \vee \langle d_i \rangle \leq \langle x_i \rangle \vee d$, but

$$\langle x_i \rangle \vee \langle d_i \rangle = \langle x_i \vee d_i \rangle = \langle 1 \rangle,$$

that is, $\langle 1 \rangle \leq \langle x_i \rangle \vee d$ and therefore $\langle x_i \rangle \vee d = \langle 1 \rangle$. By above claim, $\bigoplus_{i\in I} A_i$ is tidy. $\square$

Corollary 4.5. *The subcategory of tidy frames is a coreflective subcategory of* Frm.

5. Stacked and strongly stacked frames

Lemma 5.1. *If* $\alpha \in \mathrm{Ord}$ *and* $X \in CS$, *then*

$$\partial_F^\alpha(X) = \big(f_F^\alpha(X')\big)'. \tag{6}$$

Proof. By transfinite induction on $\alpha \in \mathrm{Ord}$:

- If $\alpha = 0$, then $\partial_F^0 = \mathrm{id}$, that is, $\partial_F^0(X) = X$ and $f_F^0 = \mathrm{id}$. Thus
$$\partial_F^0(X) = X = (X'') = \big(f_F^0(X')\big)'.$$
- Assume that $\partial_F^\alpha(X) = \big(f_F^\alpha(X')\big)'$ holds. By definition,
$$\partial_F^{\alpha+1}(X) = \partial_F\big(\partial_F^\alpha(X)\big).$$
Furthermore,
$$\partial_F^{\alpha+1}(X) = (f_F((\partial_F^\alpha(X))'))'.$$

By induction hypothesis, $\partial_F^\alpha(X) = \big(f_F^\alpha(X')\big)'$ and substituting we obtain

$$\partial_F^{\alpha+1}(X) = (f_F(f_F^\alpha(X')))' = \big(f_F^{\alpha+1}(X')\big)'$$

which is what we wanted.

- Let λ be a limit ordinal and assume that the statement holds for all $\alpha < \lambda$. By definition,

$$\partial_F^\lambda(X) = \bigcap_{\alpha<\lambda} \partial_F^\alpha(X).$$

By De Morgan laws, we have that

$$\big(\partial_F^\lambda(X)\big)' = \Big(\bigcap_{\alpha<\lambda} \partial_F^\alpha(X)\Big)' = \bigcup_{\alpha<\lambda} \big(\partial_F^\alpha(X)\big)'.$$

Using the induction hypothesis, for all $\alpha < \lambda$,

$$\big(\partial_F^\alpha(X)\big)' = f_F^\alpha(X'),$$

Then,

$$\big(\partial_F^\lambda(X)\big)' = \bigcup_{\alpha<\lambda} f_F^\alpha(X') = f_F^\lambda(X').$$

Therefore,

$$\big(\partial_F^\lambda(X)\big)' = f_F^\lambda(X'),$$

that is,

$$\partial_F^\lambda(X) = \big(f_F^\lambda(X')\big)'.$$

$\square$

As a particular case, we have the following.

Corollary 5.2. *If $X \in \mathcal{C}S$, $U \in \mathcal{O}S$ and $\alpha = \infty$, then*

$$\begin{aligned} \partial_F^\infty(X) = (v_F(X'))' &\iff \partial_F^\infty(U')' = (v_F(U)) \\ &\iff \partial_F^\infty(X)' = (v_F(X')). \end{aligned} \tag{7}$$

Lemma 5.3. *If $A = \mathcal{O}S$ and $E \subseteq S$, then*

$$\mathcal{U}_*[E]\mathcal{U} = [E].$$

Proof. Since A is spacial, then $\mathcal{U}$ is an isomorphism. Thus, if $U \in \mathcal{O}\,\mathrm{pt}\,A$ exists $V \in \mathrm{pt}\,A$ such that $U = \mathcal{U}(V)$ and as $U \in \mathrm{pt}\,A$, then $U = V$. Next,

$$\begin{aligned} (\mathcal{U}_*[E]\mathcal{U})(U) &= \mathcal{U}_*[E](\mathcal{U}(U)) = \mathcal{U}_*([E](U)) \\ &= \mathcal{U}_*(E \cup U)^\circ = \bigcup\{W \in \mathcal{O}S \mid W \subseteq (E \cup U)^\circ\} \\ &= (E \cup U)^\circ = [E](U). \end{aligned}$$

Therefore $\mathcal{U}_*[E]\mathcal{U} = [E]$. $\square$

Proposition 5.4. . *For $F \in A^\wedge$, $Q \in \mathcal{Q}S$ and $\nabla = \nabla(Q)$, if $k \in [v_\nabla, w_\nabla]$, then $\mathcal{U}_* \circ k \circ \mathcal{U} \in [v_F, w_F]$.*

Proof. If $k \in [v_\nabla, w_\nabla]$, then $\mathcal{U}_* \circ k \circ \mathcal{U} \in NA$. Additionally, for $F \in A^\wedge$ if $j \in [v_F, w_F]$, then $F = \nabla(j)$. In this way we must prove that $F = \nabla(\mathcal{U}_* \circ k \circ \mathcal{U})$.

By Hofmann-Mislove Theorem, we have that

$$x \in F \iff Q \subseteq \mathcal{U}(x) \iff \mathcal{U}(x) \in \nabla = \nabla(Q).$$

Furthermore, for all $x \in A$

$$(\mathcal{U}_* \circ k \circ \mathcal{U})(x) = \mathcal{U}_*(k(\mathcal{U}(x))) = \bigwedge(S \setminus k(\mathcal{U}(x))).$$

Since $k \in [v_\nabla, w_\nabla]$, then $\nabla = \nabla(k)$, that is, for $V \in \nabla$ it holds that $k(V) = S$. Therefore,

$$\begin{aligned} x \in F \iff \mathcal{U}(x) \in \nabla &\iff S \setminus k(\mathcal{U}(x)) = \emptyset \\ &\iff \bigwedge(S \setminus k(\mathcal{U}(x))) = 1 \\ &\iff x \in \nabla(\mathcal{U}_* \circ k \circ \mathcal{U}) \end{aligned}$$

□

We have a morphism $\mho\colon [v_\nabla, w_\nabla] \to [v_F, w_F]$. Note that this occurs at both ends of the interval on NA.

Take $k \in [v_\nabla, w_\nabla]$. Since v_F is a fitted nuclei, then it holds

$$v_F \leq \mathcal{U}_* \circ k \circ \mathcal{U}. \tag{8}$$

Similarly, as w_F is the largest element in its block,

$$\mathcal{U}_* \circ k \circ \mathcal{U} \leq w_F. \tag{9}$$

Proposition 5.5. *If $F \in A^\wedge$ and $\nabla \in \mathcal{O}S^\wedge$, then*

$$w_F = \mathcal{U}_* \circ w_\nabla \circ \mathcal{U}.$$

Proof. By (9), we must verify that $w_F \leq \mathcal{U}_* \circ w_\nabla \circ \mathcal{U}$. Equivalently, we must prove that

$$\mathcal{U} \circ w_F \leq w_\nabla \circ \mathcal{U},$$

that is, for all $x \in A$

$$\mathcal{U}(w_F(x)) \subseteq [M'](\mathcal{U}(x)) = (M' \cup \mathcal{U}(x))^\circ.$$

Since $\mathcal{U}(w_F(x)) \in \mathcal{O}S$, it is enough that $\mathcal{U}(w_F(x)) \subseteq M' \cup \mathcal{U}(x)$. As a contradiction, let $y \in \mathcal{U}(w_F(x))$ be such that $y \notin M'$ and $y \notin \mathcal{U}(x)$. Then

$$y \in M \quad \text{and} \quad x \leq y.$$

By hypothesis, $y \in \mathcal{U}(w_F(x))$, then

$$w_F(x) \nleq y \quad \text{and} \quad w_F(x) = \bigwedge\{m \in M \mid x \leq m\},$$

that is, $w_F(x) \leq y$ and it is a contradiction.

Therefore, $\mathcal{U}(w_F(x)) \subseteq M' \cup \mathcal{U}(x)$. □

Definition 5.6. Let A be a frame.

(1) A is *strongly stacked* if $\mathcal{U}_*[Q']\mathcal{U} = v_F$.
(2) A is *stacked* if $(\mathcal{U}_*([Q'])\mathcal{U})(0) = v_F(0)$.

for any (F, Q) HM pair.

Note that strongly stacked implies stacked. Our aim is to prove that the above properties are conservative, that is, S is a stacked (strongly stacked) space if and only if its frame of open sets $\mathcal{O}S$ is stacked (strongly stacked).

Proposition 5.7. *The notions of strongly stacked and stacked are conservative.*

Proof. Consider $S \in \text{Top}$, $A = \mathcal{O}S$, $F \in A^\wedge$ and $Q \in \mathcal{Q}S$.

First, suposse that S is strongly stacked, then $v_F = [Q']$ and, by Lemma 5.3, $\mathcal{U}_*[Q']\mathcal{U} = [Q'] = v_F$. Therefore $\mathcal{O}S$ is strongly stacked.

Now, suppose that S is stacked, then

$$\overline{Q} = \partial_F^\infty(S) \iff \overline{Q}' = (\partial_F^\infty(S))'.$$

Furthermore,

$$\begin{aligned}
(\mathcal{U}_*[Q']\mathcal{U})(\emptyset) &= \mathcal{U}_*([Q'](\{V \in \operatorname{pt}\mathcal{O}S \mid \emptyset \not\subseteq V\})) \\
&= \mathcal{U}_*([Q'](\emptyset)) = \mathcal{U}_*(Q')^\circ \\
&= \bigcup\{W \in \mathcal{O}S \mid W \subseteq (Q')^\circ\} \\
&= (Q')^\circ = \overline{Q}'.
\end{aligned}$$

By Corollary 5.2, $v_F(\emptyset) = (\partial_F^\infty(\emptyset'))'$. Therefore,

$$v_F(\emptyset) = (\mathcal{U}_*[Q']\mathcal{U})(\emptyset),$$

and $\mathcal{O}S$ is stacked.

$\square$

6. Separation properties in frames

If $S \in \text{Top}$ is an T_1 and strongly stacked, then it holds that $w_\nabla = v_\nabla$. Additionally, if S is stacked and T_1, then it is sober.

Thus, we could ask ourselves if the same occurs in frames. In this section we will study the above question.

Take $p \in \operatorname{pt} A$ and $(F_p, \uparrow p)$ a HM pair, then for

$$M = \{m \in A \setminus F_p \mid (\forall a \in A \setminus F_p)[a \leq m]\},$$

the nuclei $w_{F_p}(x) = \bigwedge\{m \in M \mid x \leq m\}$ is $w_p(x)$.

To prove this, note that $A \setminus F_p = \{x \in A \mid x \leq p\} = \downarrow p$. As,

$$M = \{m \in \downarrow p \mid [\forall a \in \downarrow p](a \leq m)\} = \{p\}.$$

Hence, $w_{F_p}(x) = \bigwedge\{p \mid x \leq p\}$, that is,

$$w_{F_p}(x) = \begin{cases} 1, & \text{if} \quad x \nleq p \\ p, & \text{if} \quad x \leq p \end{cases} = w_p(x) \tag{10}$$

for all $x \in A$. Therefore $w_{F_p} = w_p$.

Proposition 6.1. *Let $A \in \mathrm{Frm}$ and (F, Q) be any HM pair. If $v_F(0) = w_F(0)$, then for all $p \in \mathrm{pt}\, A$, p is maximal in $\mathrm{pt}\, A$.*

Proof. Consider $p, q \in \mathrm{pt}\, A$ such that $p \leq q$. Note that for $q \in \mathrm{pt}\, A$, we have $(F_q, \uparrow q)$ a HM pair, with

$$F_q = \{x \in A \mid \uparrow q \subseteq \mathcal{U}(x)\} = \{x \in A \mid (\forall r \in \mathrm{pt}\, A)[r \leq q \Rightarrow x \nleq q]\}.$$

By hypothesis, $v_{F_q}(0) = w_{F_q}(0)$ and by (10)

$$q = w_q(0) = w_{F_q}(0) = v_{F_q}(0) \leq v_{F_q}(p) = p.$$

Therefore, $p = q$ and every point is maximal in $\mathrm{pt}\, A$. $\square$

As we show later on (2), for all (F, Q, ∇) HM tupla we have

$$\mathcal{U} \circ v_F \leq v_\nabla \circ \mathcal{U}.$$

Additionally, if A is strongly stacked, then $v_F = \mathcal{U}_*[Q']\mathcal{U}$ and

$$\mathcal{U}_* \circ v_\nabla \circ \mathcal{U} \leq \mathcal{U}_* \circ [Q']\mathcal{U}$$

Thus, $v_F = \mathcal{U}_* \circ v_\nabla \circ \mathcal{U}$, equivalently $\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$.

Therefore, if A is strongly stacked, then the diagram (2) is commutative. The following result tells us under which conditions we have the converse implication.

Proposition 6.2. *If $A \in \mathrm{Frm}$ is tidy and the diagram*

$$\begin{tikzcd} A \arrow[r, "v_F"] \arrow[d, "\mathcal{U}"'] & A \arrow[d, "\mathcal{U}"] \\ \mathcal{O}S \arrow[r, "v_\nabla"'] & \mathcal{O}S \end{tikzcd}$$

commutes, then A is strongly stacked.

Proof. Let $A \in \mathrm{Frm}$, $S = \mathrm{pt}\, A$ and (F, Q) be any HM pair. By hypothesis, A is tidy and by previous results we have the implications

$$A \text{ tidy } \Rightarrow \mathcal{O}S \text{ tidy } \Rightarrow S \text{ stacked and packed.}$$

Since S is packed, if $Q \in \mathcal{Q}S$, then $Q \in \mathcal{C}S$ and $Q' \in \mathcal{O}S$. Then, as S is stacked, we have that $\overline{Q} = \partial_\nabla^\infty(S)$, that is, $Q' = v_\nabla(\emptyset)$ and as $\mathcal{O}S$ is tidy, $v_\nabla = u_{Q'} = [Q']$.

Additionally, if $v_\nabla \circ \mathcal{U} = \mathcal{U} \circ v_F$, then

$$v_\nabla \circ \mathcal{U} \leq \mathcal{U} \circ v_F \quad \text{ and } \quad \mathcal{U} \circ v_F \leq v_\nabla \circ \mathcal{U},$$

that is,

$$[Q'] \circ \mathcal{U} \leq \mathcal{U} \circ v_F \quad \text{and} \quad \mathcal{U} \circ v_F \leq [Q'] \circ \mathcal{U}.$$

By properties of the right adjoint

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} \leq v_F \quad \text{and} \quad v_F \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Therefore $v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}$ and A is strongly stacked. □

Consequently, we have the following result.

Corollary 6.3. *If A is tidy, then $S = \operatorname{pt} A$ is strongly stacked.*

Proof. Since S is strongly stacked if and only if $v_\nabla = [Q']$, the result follows from the fact that

$$\mathcal{O}S \text{ tidy } \Rightarrow S \text{ strongly stacked}$$

□

With the morphism $\mathcal{U}_A$, we have the nuclei sp_A defined by

$$\operatorname{sp}_A(x) = \bigwedge\{p \in \operatorname{pt} A \mid x \leq p\}$$

for $x \in A$. Combining this fact with the Proposition 6.1 we know information about the collapse on admisibility interval.

Proposition 6.4. *For $A \in \mathrm{Frm}$ and (F, Q) any HM pair, the following statements hold:*

1. *If A is T_1 and stacked, then $v_F(0) = w_F(0)$.*
2. *If $v_F(0) = w_F(0)$, then A is stacked.*
3. *If $v_F(0) = w_F(0)$ and the set $Y = \{q \in \operatorname{pt} A \mid a \leq q\}$ is non-empty for all $a \in A$, then A is T_1.*

Proof.

(1) If A is T_1, then

$$\mathcal{O}S \text{ is } T_1 \Rightarrow S \text{ is } T_1 \Rightarrow Q = M.$$

Since A is stacked,

$$v_F(0) = (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) = (\mathcal{U}_* \circ [M'] \circ \mathcal{U})(0) = w_F(0).$$

(2) Recall that for every HM pair, the following inequality holds:

(11) $$v_F \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U} \leq w_F$$

and by hypothesis $v_F(0) = w_F(0)$, that is, $v_F(0) = (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) = w_F(0)$. Therefore, A is stacked.

(3) Consider $p \in \operatorname{pt} A$ and $a \in A$ such that $p \leq a$. By hypothesis $\{q \in \operatorname{pt} A \mid a \leq q\} \neq \emptyset$. Then

$$p \leq a \leq \operatorname{sp}(a) = \bigwedge q \in \operatorname{pt} A \mid a \leq q \leq q$$

for some $q \in Y$. Thus, $p \leq a \leq q$ and by Proposición 6.1, $p = q$. Therefore, $p = a$ and every point is maximal in A, that is, A is T_1.

□

Corollary 6.5. *For $A \in \mathrm{Frm}$ and (F, Q) any HM pair, the following statements hold:*

1. *If A is T_1 and strongly stacked, then $v_F = w_F$.*
2. *If $v_F = w_F$, then A is strongly stacked.*
3. *If $v_F = w_F$ and the set $Y = \{q \in \mathrm{pt}\, A \mid a \leq q\}$ is non-empty for all $a \in A$, then A is T_1.*

Proof.

1. If A is strongly stacked, then $v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}$ and by T_1 we have that $Q = M$. Therefore $v_F = w_F$.
2. This follows from (11).
3. This follows from Proposition 6.4.

$\square$

Note that, by 6.1 and 6.4, if $v_F(0) = w_F(0)$, then A is stacked and every point is maximal in $\mathrm{pt}\, A$. Particularly, for each $p \in \mathrm{pt}\, A$ and $(F_p, \uparrow p)$ it holds $v_{F_p}(0) = p$. Furthermore,

$$v_{F_p}(p) = v_{F_p}(0) = 0_{A_{F_p}}.$$

If A is T_1, p is maximal in A, then p is maximal in A_{F_p} and $A_{F_p} = \{q, 1\}$.

Now, $F_{F_p} = \nabla(v_{F_p})$ and $x \in F_p$ if and only if $v_{F_p}(x) = 1$. In this way, under the previous hypotheses, it holds that

$$v_{F_p}(x) = \begin{cases} 1, & \text{si} \quad x \nleq p \\ \\ p, & \text{si} \quad x \leq p \end{cases} = \begin{cases} 1, & \text{si} \quad x \in F_p \\ \\ p, & \text{si} \quad x \notin F_p \end{cases} = w_{F_p}(x).$$

Therefore, if A is T_1 and stacked, then $v_{F_p} = w_{F_p}$ for each $p \in \mathrm{pt}\, A$. With this in mind, we could ask ourselves if there exists some way to recover the notion of strongly stacked by means of T_1 and stacked.

Now, we need the following information:

1. Let $A^\wedge$ be a topological space where the basic open sets of $\mathcal{O}(A^\wedge)$ are given by
$$D(x) = \{F \in A^\wedge \mid x \in F\}$$
where $x \in A$.
2. For $A \in \mathrm{Frm}$, we can construct a frame VA as in [Sim04]. Each element in $\mathrm{pt}(VA)$ is of the form
$$(F, a), \quad \text{where } F \in A^\wedge \text{ y } a \in [v_F(0), w_F(0)].$$
3. Take a continuos surjective map $\pi\colon \mathrm{pt}(VA) \to A^\wedge$, defined by
$$(F, a) \mapsto F.$$
4. By the functor $\mathcal{O}(-)$ we have
$$\mathcal{O}(\pi) = \pi^{-1}\colon \mathcal{O}(A^\wedge) \to \mathcal{O}(\mathrm{pt}(VA)).$$

(5) Open sets in $\mathrm{pt}(VA)$ are given by the following:

$$(F,a) \in [-](x) \iff x \in F \quad \text{and} \quad (F,a) \in \langle - \rangle(x) \iff x \nleq a. \tag{12}$$

Particularly,

$$a \in [F](x) \iff x \in F \quad \text{and} \quad a \in \langle F \rangle(x) \iff x \nleq a. \tag{13}$$

Proposition 6.6. *Let $A, VA \in \mathrm{Frm}$ and $A^\wedge$ be as above. The following statements hold:*

(1) *If A is T_1 and stacked, then $\mathcal{O}(A^\wedge) \cong \mathcal{O}(\mathrm{pt}(VA))$ under the continuous function π.*
(2) *If the morphism $\mathcal{O}(\pi) = \pi^{-1}$ induces that $\mathcal{O}(A^\wedge) \cong \mathcal{O}(\mathrm{pt}(VA))$, then A is stacked.*
(3) *If the morphism $\mathcal{O}(\pi) = \pi^{-1}$ induces that $\mathcal{O}(A^\wedge) \cong \mathcal{O}(\mathrm{pt}(VA))$ and*

$$\mathcal{V}(a) = \{q \in \mathrm{pt}\, A \mid a \leq q\} \neq \emptyset,$$

then A is T_1.

Proof.

(1) If A is T_1 and stacked, then for each $F \in A^\wedge$ it holds that

$$v_F(0) = w_F(0).$$

Take $(F,a), (G,b) \in \mathrm{pt}(VA)$ such that $\pi(F,a) = F = \pi(G,a) = G$. By T_1 and stacked we have

$$a = v_F(0) = v_G(0) = b.$$

Therefore $(F,a) = (G,b)$, that is, π is injective. Since it is also surjective, π is bijective.

We must show that $\pi^{-1} \colon A^\wedge \to \mathrm{pt}(VA)$ is continuous. For this, recall that by (12)

$$\begin{aligned} (F,a) \in [-](x) &\iff x \in F \\ &\iff (F,a) \in \pi^{-1}(D(x)) \\ &\iff \pi([-](x)) = D(x). \end{aligned}$$

For the other basic open sets, by (13) we have

$$a \notin \langle F \rangle(x) \iff x \leq a \iff v_F(x) \leq a.$$

By T_1 and stacked we have that $a = v_F(0)$ and thus

$$v_F(0) \notin \langle F \rangle(x) \iff x \leq v_F(0) \iff v_F(x) = v_F(0).$$

As $v_F(0) \neq 1$,

$$v_F(0) \notin \langle F \rangle(x) \iff x \notin F$$

or equivalently

$$a \in \langle F \rangle(x) \iff (F,a) \in \langle - \rangle(x) \iff x \in F.$$

Therefore, $\pi(\langle - \rangle(x)) = D(x)$ y π^{-1} is continuous. Hence, π is a homeomorphism and

$$\mathcal{O}(A^\wedge) \cong \mathcal{O}(\mathrm{pt}(VA)).$$

(2) Suppose that $\mathcal{O}(A^\wedge) \cong \mathcal{O}(\mathrm{pt}(VA))$ under $\mathcal{O}(\pi) = \pi^{-1}$. Since π^{-1} is bijective, then π is also bijective. Thus, if $F \in A^\wedge$, then

$$(F, v_F(0)), (F, w_F(0)) \in \mathrm{pt}(VA)$$

and

$$\pi(F, v_F(0)) = F = \pi(F, w_F(0)).$$

By the injectivity of π we have that $v_F(0) = w_F(0)$ and therefore A is stacked.

(3) Suppose that $\mathcal{O}(A^\wedge) \cong \mathcal{O}(\mathrm{pt}(VA))$ and A is stacked. Consider $a \in A$ and $p \in \mathrm{pt}\, A$ such that $p \leq a$. Moreover, by the previous item we have that

$$\mathcal{O}(A^\wedge) \cong \mathcal{O}(\mathrm{pt}(VA)) \Rightarrow v_F(0) = w_F(0)$$

and every point is maximal in $\mathrm{pt}\, A$. Since $\mathcal{V}(a) \neq \emptyset$, there exists $q \in \mathrm{pt}\, A$ such that $a \leq q$. By the maximality of p and q, we have

$$p \leq a \leq q = p.$$

Therefore, $p = a$ and A is T_1.

$\square$

Note that item (3) holds whenever A is spatial. In the spatial setting, this corresponds to the topological fact that

$$S \text{ is } T_1 \text{ and stacked} \Rightarrow S \text{ is sober.}$$

However, the point-free setting allows for a potentially broader phenomenon, since the conditions of being T_1 and stacked do not a priori force a frame to be spatial. This leads naturally to the following question:

Do there exist T_1 stacked frames that are not spatial?

7. Quotients of stacked frames

Remember that for $(f\colon A \to B) \in \mathrm{Frm}$, $j \in NA$ and $k \in NB$, we have the following definitions:

$$N(f)(j) = \bigvee \{ u_{f(j(a))} \wedge v_{f(a)} \mid a \in A \} \in NA_j$$

and

$$N(f)_*(k) = f_* \circ k \circ f \in NA,$$

where f_* is the right adjoint of f.

Proposition 7.1. *Let A be a frame and $j \in NA$. If A is strongly stacked, then A_j is strongly stacked.*

Proof. Consider $S = \operatorname{pt} A$, $S_j = \operatorname{pt} A_j$ and let $F \in A_j^\wedge$ be. We define:

$$G = j_*(F), \quad Q = S_j \setminus F, \quad R = S \setminus G,$$

where j_* is the right adjoint associated with the frame morphism $j^*\colon A \to A_j$ and $x \in G$ if and only if $j(x) \in F$. Considering the spatial reflections

$$\mathcal{U}\colon A \to \mathcal{O}S \quad \text{and} \quad \mathcal{U}_j\colon A_j \to \mathcal{O}S_j$$

we have the following diagram

$$\begin{tikzcd}
 & & A \arrow[r, "\mathcal{U}"] \arrow[d, "j^*"'] & \mathcal{O}S \arrow[d, "sj"] \\
A \arrow[urr, "v_G"] \arrow[r, "\mathcal{U}"] \arrow[d, "j^*"'] & \mathcal{O}S \arrow[urr, "{[R']}"] \arrow[d, "sj"] & A_j \arrow[r, "\mathcal{U}_j"'] & \mathcal{O}S_j \\
A_j \arrow[urr, "v_F"] \arrow[r, "\mathcal{U}_j"'] & \mathcal{O}S_j \arrow[urr, "{[Q']}"'] & &
\end{tikzcd}$$

Using the right adjoints of $\mathcal{U}$ and $\mathcal{U}_j$ we obtain

$$v_F, \quad l = N(\mathcal{U}_j)_*([Q']), \quad m = N(j^*)(N(\mathcal{U})_*)([R']), \quad n = N(j^*)(N(\mathcal{U}_j \circ j^*)_*)([Q'])$$

are nuclei in NA_j. Let us see what is the relationship between these nuclei.

By construction, we have $v_F \leq N(\mathcal{U}_j)_*([Q'])$ and we must verify the other inequality for A_j to be strongly stacked.

By hypothesis, we have $v_G = N(\mathcal{U})_*([R'])$. Moreover, for $x \in A$, we have $j(v_x) \leq v_{j(x)}$ and, by definition of the adjusted nuclei associated with F and G,

$$j \circ v_G \leq v_F. \tag{14}$$

Furthermore, for $(j\colon A \to A_j) \in \mathrm{Frm}$

$$N(j^*)(v_G) = \bigvee\{u_{j(v_G(a))} \wedge v_{j(a)} \mid a \in A\} \quad \text{y} \quad v_F = \bigvee\{u_{v_F(a)} \wedge v_a \mid a \in A_j\}.$$

Note, by (14), $u_{j(v_G)} \leq u_{v_F}$ and therefore

$$m \leq v_F. \tag{15}$$

To l and n, we have

$$n = \bigvee\{u_{j^*(n(x))} \wedge v_{j^*(x)} \mid x \in A\} \quad \text{and} \quad l = \bigvee\{u_{l(x)} \wedge v_x \mid x \in A_j\},$$

where

$$j^*(n(x)) = j^*((j_* \circ \mathcal{U}_{j*} \circ [Q'] \circ U_j \circ j^*)(x)) \quad \text{y} \quad l(x) = N(\mathcal{U}_j)_*([Q'])(x).$$

Then,

$$\begin{aligned} j^*(n(x)) &= j^*(j_*(\bigvee\{a \in A \mid \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(j(x))\})) \\ &= \bigvee\{a \in A \mid \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(j(x))\}. \end{aligned}$$

Since $x \in A_j$,

$$l(x) = \bigvee\{a \in A \mid \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(j(x))\}.$$

Therefore $j(n(x)) = l(x)$.

To $m(x)$ we have

$$m(x) = \bigvee\{u_{j^*(\mathcal{U}_* \circ [R'] \circ \mathcal{U})(x)} \wedge v_{j^*(x)} \mid x \in A\}.$$

Since $A_j \subseteq A$, then $Q = R \cap A_j$ and

$$m(x) = \bigvee\{u_{j^*(\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(x)} \wedge v_x \mid x \in A_j\}.$$

Note

$$\begin{aligned} j^*(\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(x) &= j^*(\bigvee\{a \in A \mid \mathcal{U}(a) \subseteq Q' \cup \mathcal{U}(x)\}) \\ &= \bigvee\{j^*(a) \in A_j \mid \mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(x)\}. \end{aligned}$$

To $x \in A_j$

$$\begin{aligned} \mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(j^*(x)) &\iff \mathcal{U}(j^*(a)) \cap S_j \subseteq (Q' \cup \mathcal{U}(j^*(x))) \cap S_j \\ &\iff \mathcal{U}_j(j^*(a)) \subseteq Q' \cup \mathcal{U}_j(j^*(x)), \end{aligned}$$

that is,

$$\bigvee\{j^*(a) \in A_j \mid \mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(x)\} = j^*(N(\mathcal{U}_j \circ j)_*)([Q']).$$

Thus, we have

$$\bigvee\{u_{j^*(N(\mathcal{U}_j \circ j)_*)[Q'](x)} \wedge x \mid x \in A\} = m(x).$$

Since $(N(\mathcal{U}_j \circ j)_*)([Q']) \leq j(N(\mathcal{U}_j \circ j)_*)([Q'])$ and by (15)

$$l \leq m \leq v_F \leq l.$$

Therefore $v_F = \mathcal{U}_{j*} \circ [Q'] \circ \mathcal{U}_j$, that is, A_j is strongly stacked. □

Since every strongly stacked frame is stacked, we obtain the following result.

Corollary 7.2. *Let A be a frame and $j \in NA$. If A is strongly stacked, then A_j is stacked.*

Proof. By the previous proposition, if A is strongly stacked, then A_j is strongly stacked and this implies that A_j is stacked. □

8. Relationship to the property (H)

If A is a Hausdorff frame, then A is T_1. In this way, to be able to use the example of Paseka and Smarda it is enough that $(\mathbf{H})$ implies stacked.

Proposition 8.1. *If $A \in \mathrm{Frm}$ satisfies $(\mathbf{H})$, then A is stacked.*

Proof. If A satisfies $(\mathbf{H})$, then A is T_1, that is, for every pair of points $x, y \in A$ with $x \neq y$, there exist neighborhoods U, V of x, y respectively such that $U \cap V = \emptyset$. This implies that

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = w_F,$$

that is, $\mathcal{U}_* \circ [Q'] \circ \mathcal{U}(0) = w_F(0)$ and since $w_F \neq \mathrm{tp}$, we have $w_F(0) \neq 1$.

By construction, we have that $v_F(0) \leq w_F(0)$ and thus, we need to verify that the reverse inequality holds to conclude that A is stacked. For a contradiction, suppose that

$$w_F(0) \nleq v_F(0).$$

Since $1 \neq w_F(0) \nleq v_F(0)$, by $(\mathbf{H})$, exists $c \in A$ such that

$$c \nleq w_F(0) \quad \text{y} \quad \neg c \nleq v_F(0).$$

If $\neg c \nleq v_F(0)$, then $c \notin F$ □

9. Marcos KC

Definition 9.1. Let A be a frame. We say that A is:

(1) KC if each compact quotient is closed.
(2) *Hausdorff closed compact* (or KCH for short) if each compact quotient of A is closed and Hausdorff.

La Definición 9.1 nos dice que si $j \in NA$, entonces $\nabla(j) \in A^\wedge$ y $j = u_a$ para algún $a \in A$. De manera adicional, para 2) pedimos que A_j cumpla la propiedad $(\mathbf{H})$.

Proposition 9.2. *If $A \in \mathrm{Frm}$ satisfies KC, then A_j satisfies KC for each $j \in N(A)$.*

Proof. Consider $k \in NA_j$ such that $(A_j)_k$ is compact. If $(A_j)_k$ is compact, then $\nabla(k) \in A_j^\wedge$ and, by Proposition 3.1, $j_*\nabla(K) \in A^\wedge$.

Let $l = j_* \circ k \circ j^* \in NA$ be, then A_l is a compact quotient of A and there exists $a \in A$ such that $l = u_a$. Thus

$$\begin{tikzcd} A \arrow[r, "j^*"'] \arrow[rrr, bend left, "l"] & A_j \arrow[r, "k"'] & (A_j)_k \arrow[r, "j_*"'] & A_j \subseteq A \end{tikzcd}$$

and $a \vee x = k(j(x))$. Therefore, if $x = a$, $k(j(x)) = a$.

We need that $k = u_b$ for some $b \in A_j$. If $x \in A_j$ and $b = j(a)$

$$\begin{aligned} u_b(x) = b \vee x = b \vee j(x) &= j(j(a) \vee j(x)) \\ &= j(k(j(a)) \vee x) \\ &= j(u_a(x)) \\ &= j(k(x)) \\ &= k(x). \end{aligned}$$

Hence $u_b = k$. □

Proposition 9.3. *If A is a KC frame, then A is a T_1 frame.*

Proof. Let $p \in \operatorname{pt} A$ and $a \in A$ such that $p \leq a \leq 1$. Consider

$$w_p(x) = \begin{cases} 1 & \text{si} \quad x \nleq p \\ \\ p & \text{si} \quad x \leq p \end{cases}$$

to $x \in A$. $P = \nabla(w_p) = \{x \in A \mid x \nleq p\}$ is a completely prime filter (in particular, $P \in A^\wedge$). Becuase A is KC, then A_{w_p} is a compact closed quotient. Thus $u_p = w_p$ and

$$u_p(a) = a \quad \text{y} \quad w_p(a) = 1.$$

that is, $a = 1$. Therefore p is maximal. □

One of our aims is study nuclei such that produce compact quotient, particularly, nuclei that produce compact closed quotient.

Define the following subset of A

$$uA = \{a \in A \mid A_{u_a} \text{ es un cociente compacto}\}.$$

Note that, to $a \in uA$ we have $\nabla(u_a) \in A^\wedge$, that is, define the map

$$\alpha\colon uA \to A^\wedge.$$

Let us see the properties of this map.

Lemma 9.4. *For $uA \subseteq A$ as before, the following statements hold:*

(1) *If A is tidy, then α is surjective.*
(2) *If u_a is fitted for all $a \in uA$, then α is injective.*
(3) *If A is tidy and $a \in uA$, then α is bijective if and only if u_a is fitted.*

Proof.

(1) Consider $F = \nabla(j) \in A^\wedge$ and v_F the fitted nucleus associated with F. If A is tidy, then $v_F = u_d$, where $d = v_F(0) \in A$. Hence,

$$\alpha(d) = \nabla(u_d) = \nabla(v_F) = F.$$

and α is surjective.

(2) Consider $a, b \in uA$ such that $\alpha(a) = \alpha(b)$, that is, $\nabla(u_a) = \nabla(u_b)$. By hypothesis, u_a, u_b are fitted nuclei, that is,

$$u_c \leq u_d \quad \text{y} \quad u_c \geq u_d.$$

Evaluating both nuclei at 0 we get $c = d$ and therefore α is injective.

(3) Let A be tidy, $a \in uA$ and suppose that α is bijective. Consider $F = \nabla(u_a)$ and v_F the fitted nucleus associated with F. By efficiency, $v_F = u_d$, that is,

$$\alpha(a) = \nabla(u_a) = \nabla(v_F) = \nabla(u_d) = \alpha(d)$$

where $d = v_F(0)$.

By hypothesis, α is bijective, then, by injectivity, $a = d$. Therefore $v_F = u_a$ and u_a is fitted.

Conversely, since A is tidy, then by 1), α is surjective. By hypothesis, u_a is fitted and, by 2), α is injective. Therefore α is bijective.

□

Using KC frames, we have the following result.

Corollary 9.5. *Let A be a KC frame and $F \in A^\wedge$. Then α is bijective if and only if for all $j \in [v_F, w_F]$, j is fitted.*

Proof. Suppose that A is a KC frame and let $F \in A^\wedge$. By definition, for all $j \in [v_F, w_F]$, we have $j = u_a$ for some $a \in A$. Since A is a KC frame, it is efficient and by the previous lemma, α is bijective if and only if for all $j \in [v_F, w_F]$, j is fitted. □

Proposition 9.6. *Let A be a frame. A is KC if and only if A is tidy and $\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$.*

Proof.

$\Rightarrow$)**:** Suppose that A is a KC frame. Trivially, KC implies efficient and thus, it remains to verify that KC implies that $\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$ or equivalently, by properties of the right adjoint,

$$v_F = \mathcal{U}_* \circ v_\nabla \circ \mathcal{U}.$$

If A is a KC frame, then for any $F \in A^\wedge$, it holds that for all $j \in [v_F, w_F]$, $j = u_a$ for some $a \in A$. In particular,

$$v_F = u_d, \quad \text{y} \quad \mathcal{U}_* \circ v_\nabla \circ \mathcal{U} = u_a$$

where $d, a \in A$.

By construction, always holds that $u_d \leq u_a$. To see the other inequality, we verify that the function α is injective. Note that $d \leq a$

$\Leftarrow$)**:** Suppose that A is efficient and $\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$ for all HM triples (F, Q, ∇). First, if A is efficient, then A is T_1. Moreover, if A is efficient and $\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$, then A is strongly stacked. Therefore, A is T_1

and strongly stacked. The above implies that $v_F = w_F$ and, by efficiency, $v_F = u_d$, where $d = v_F(0)$. Thus, for all $j \in [v_F, w_F]$, $j = u_d$, meaning that A is KC.

□

10. Examples

10.1. **The frame $[0,1]$.** Following the idea presented in [Sim06b] (or, alternatively, in [Sim04]), let us consider the linearly ordered frame

$$A = [0,1].$$

In this case the following observations hold.

(1) For all $a, b \in A$ the implication $(a \succ b)$ is given by

$$(a \succ b) = \begin{cases} 1, & \text{si } a \leq b, \\ b, & \text{si } a > b. \end{cases}$$

(2) For $a, x \in A$, the distinguished nuclei u_a, v_a, w_a are given by

$$u_a(x) = \begin{cases} x & \text{si } a \leq x, \\ a & \text{si } x < a, \end{cases}, \quad v_a(x) = \begin{cases} 1 & \text{si } x \geq a, \\ a & \text{si } x < a, \end{cases} \quad w_a(x) = \begin{cases} 1 & \text{si } x > a, \\ a & \text{si } x \leq a. \end{cases}$$

(3) If $F \subseteq A$ is a filter, then F is always admissible and has the form

$$F = \nabla(w_a) = (a, 1] \quad \text{o} \quad F = \nabla(v_a) = [a, 1]$$

for some $a \in A$. In particular, $F \in A^\wedge$ if and only if $F = (a, 1]$.

(4) If $F \in A^\wedge$, the adjusted nucleus associated v_F is calculated as

$$v_F(x) = l_m(x) = \begin{cases} 1 & \text{if } x > m, \\ x & \text{if } x \leq m. \end{cases} \tag{16}$$

where $l_m = v_m \wedge w_m$ and $m = \bigwedge F$.

(5) For $S = \operatorname{pt} A$ it holds that $S = [0,1)$ and the open sets in $\mathcal{O}S$ are of the form

$$\mathcal{U}_A(x) = \{p \in S \mid p < x\} = [0, x).$$

(6) The frame A is not T_1, since no $p \in S$ is maximal. Consequently, A does not satisfy any stronger separation property in Frm.

(7) For $F \in A^\wedge$ the compact saturated set associated $Q \in \mathcal{Q}S$ is of the form $Q = [0, m]$.

(8) It holds that $[Q'] = v_F$; therefore, S is strongly stacked.

(9) Since $[0, x) \in \mathcal{O}S$, then $[x, 1] \in \mathcal{C}S$ and $Q = [0, m]$. Therefore, the compact subsets are not closed; that is, S is not packed.

(10) The frame A is not efficient.

The above resume some characteristics about the frame $[0,1]$. Now, we will perform the analysis of the patch constructions of A and its point space S.

Remember that for $A \in \mathrm{Frm}$ and $S \in \mathrm{Top}$ we have

$$\mathrm{Pbase} = \{u_a \wedge v_F \mid a \in A, F \in A^\wedge\} \quad \text{y} \quad \mathrm{pbase} = \{U \cap Q' \mid U \in \mathcal{O}S, Q \in \mathcal{Q}S\}$$

where Pbase generates the patch frame of A and pbase generates the topology of the patch space of S. By (16) we have that

$$\text{Pbase} = \{u_a \wedge l_m \mid a \in [0,1], m \in [0,1)\}.$$

For $x \in A$ we have

$$(u_a \wedge l_m)(x) = u_a(x) \wedge l_m(x).$$

Let us analyze the value of this map according to the relative position of x and m.

Case 1: If $x < m$, then

$$(u_a \wedge l_m)(x) = u_a(x) \wedge x = x.$$

Case 2: If $x = m$, then

$$(u_a \wedge l_m)(x) = u_a(m) \wedge m = x.$$

Case 3: If $x > m$, then

$$(u_a \wedge l_m)(x) = u_a(x) \wedge 1 = u_a(x).$$

In the first two cases, the value of a does not affect the evaluation. Thus, the only non-trivial case occurs when $m < x$, where the value of $(u_a \wedge l_m)(x)$ depends on the relative position of a and x. We therefore distinguish the following two subcases.

Case 3.1: If $m < x < a$, then

$$(u_a \wedge l_m)(x) = a.$$

Case 3.2: If $m < a \leq x$, then

$$(u_a \wedge l_m)(x) = x.$$

From the preceding analysis, we conclude that the non-trivial generators of Pbase correspond precisely to the pairs (m, a) with $m < a$. Consequently, the frame PA is determined by the open intervals of the form (m, a), where $m \in \operatorname{pt} A$ and $a \in A$.

We now turn to the question of whether PA satisfies any separation property. To this end, we recall the relationship between the point-free patch construction and its spatial counterpart.

Let A be an arbitrary frame and let $S = \operatorname{pt} A$ be its space of points. The two patch constructions are related by the following commutative diagram:

$$\begin{tikzcd}
A \arrow[r] \arrow[d, "\mathcal{U}_A"'] & PA \arrow[r, hook] \arrow[d, "P(\mathcal{U}_A)"] & NA \arrow[d, "N(\mathcal{U}_A)"] \\
\mathcal{O}S \arrow[r] \arrow[dr, hook] & P\mathcal{O}S \arrow[r, hook] \arrow[d, "\pi"] & N\mathcal{O}S \arrow[d, "\sigma"] \\
 & \mathcal{O}^p S \arrow[r, hook] & \mathcal{O}^f S
\end{tikzcd}$$

where $\mathcal{O}^f S$ is the Skula topology (or front topology) of S. In [SS06b] is named *the patch assembly diagram* and more details about it can also be found in [Sex03].

In our case, $\mathcal{U}_A$ is an isomorphism. Moreover, under $\mathcal{U}_A$, the patch construction turns out to be functorial and, therefore, $PA \cong P\mathcal{O}S$. Similarly, $NA \cong N\mathcal{O}S$. To describe explicitly $P\mathcal{O}S$ and $N\mathcal{O}S$, we use the following result, which can be consulted in [ÁBMZ20].

Proposition 10.1. *Let S be a partially ordered set.*

(1) *If S has no infinite antichains, then $N\mathcal{O}S$ is spatial.*
(2) *If S is totally ordered, then $N\mathcal{O}S$ is spatial.*

Thus, by Proposition 10.1 we have that $N\mathcal{O}S$ is spatial and therefore $N\mathcal{O}S \cong \mathcal{O}^f S$. With all this information, the patch assembly diagram, in our particular case is as follows.

$$\begin{tikzcd}
A \arrow[r] \arrow[d, "\cong"'] & PA \arrow[r] \arrow[d, "\cong"] & NA \arrow[d, "\cong"] \\
\mathcal{O}S \arrow[r] & \mathcal{O}^p S \arrow[r] & \mathcal{O}^f S
\end{tikzcd}$$

Therefore, if we want to study the frame PA, we can study the patch topology $\mathcal{O}^p S$. Remember that

$$\text{pbase} = \{U \cap Q' \mid U \in \mathcal{O}S, Q \in \mathcal{Q}S\}.$$

In our case,

$$\text{pbase} = \{[0,a) \cap [m,1) \mid a \in [0,1], m \in [0,1)\} = \{(m,a)\}.$$

Since the topology generated by the intervals (m,a) coincides with the usual topology on $[0,1)$, the space ${}^p S$ is Hausdorff. Consequently, $\mathcal{O}^p S \cong PA$ satisfies $(\mathbf{H})$, even when A is not T_1 nor tidy.

Lemma 10.2. *Let A be a frame, if the morphism $\mathrm{p}U \colon \mathrm{p}A \to \mathrm{p}\mathcal{O}S$ is an isomorphism, then A is strongly stacked.*

Proof. Consider any open filter F on A and the compact saturated determined by it, say Q, thus we have the nuclei v_F and $U_*[Q']U = Q[U]$, we neee to observe the action of $\mathrm{p}U$ on these two nuclei, in order to do that remember that we have a conmutative diagram:

$$\begin{tikzcd}
A \arrow[r] \arrow[d, "\mathcal{U}_A"'] & \mathrm{p}A \arrow[r, hook, "\iota"] \arrow[d, "\mathrm{p}(\mathcal{U}_A)"] & NA \arrow[d, "N(\mathcal{U}_A)"] \\
\mathcal{O}S \arrow[r] \arrow[dr, hook] & \mathrm{p}\mathcal{O}S \arrow[r, hook, "\imath"'] \arrow[d, "\pi"] & N\mathcal{O}S \arrow[d, "\sigma"] \\
 & \mathcal{O}^p S \arrow[r, hook] & \mathcal{O}^f S
\end{tikzcd}$$

since $N(\mathcal{U})\iota = \imath \mathrm{p}\mathcal{U}$, that is, $\mathrm{p}\mathcal{U}$ is the restriction of $N(\mathcal{U})$ to $\mathrm{p}A$, then

$$\mathrm{p}\mathcal{U}(\mathrm{v}_F) = N(\mathcal{U})(\mathrm{v}_F) = \bigvee\{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

since the morphism $\mathcal{U}$ transforms open filters we have

$$\bigvee\{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\} = \bigvee\{u_{v_{\mathcal{U}F}} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

But this representation is the corresponding to the nucleus $\mathrm{v}_{\mathcal{U}F}$.

$\square$

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